MITOSYNC: JOINT MOTHER–DAUGHTER ASSOCIATION FOR SPARSE CELL TRACKING

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ABSTRACT

Long intervals between microscopy frames make it difficult to identify which cells are the daughters of the same mother. We present MitoSync, a cell association method that scores a mother and two candidate daughters together, then selects divisions and continuations in one optimization. The score combines image and geometry features and is unchanged when the daughters are exchanged. Trained only on real data, MitoSync improves sparse division F1 over a Trackastra graph reference by 5.24 and 7.95 percentage points on two retrospectively evaluated Cell Tracking Challenge sequences. Frozen models improve sparse event F1 by 4.05 points on 36 external LFCT fields; native-rate gains are not established. Controlled tests of programmable synthetic supervision show conditional benefits, without a consistent advantage of rendered video over direct painting. These results support joint mother–daughter association for sparse tracking and separate its contribution from the choice of synthetic training images.

***Index Terms*—** cell division, cell tracking, joint association, sparse microscopy, synthetic supervision

1. INTRODUCTION

Cell lineages describe how cells move and divide. A missed division or an incorrect daughter assignment changes the reconstructed lineage. Sparse imaging makes these decisions harder because daughters can move apart between observations. Divisions are also rare in training data [1, 2], with fewer than 7 per 1000 cell-frames in our PhC-C2DL-PSC training split.

A plausible mother–daughter link is not sufficient to identify a division. The two daughters must form a plausible pair, and their assignment must be consistent with competing continuations. MitoSync combines exchange-invariant mother–daughter scoring with Trackastra link evidence and selects child sets jointly. This real-only model is the core method; it does not require video generation.

We make three contributions: a symmetric image-and-geometry score for mother–two-daughter candidates; joint decoding in which divisions and continuations compete for cells; and controlled tests that distinguish association gains from the effects of programmed synthetic supervision. Evaluation combines two CTC development sequences with frozen-model transfer to LFCT. The controls identify conditional synthesis gains and competitive painting results.

2. RELATED WORK

LapTrack [3], MAGIK [2] and Trackastra [1] link detected cells. Zyss et al. [4] classify mother–two-daughter triplets using three-frame image and geometry features, select consistent divisions, then optimize tracks. MitoSync couples an exchange-invariant representation to Trackastra through a child-set utility: divisions and continuations compete in one optimization. We test this decoding distinction with scores fixed, without claiming a reproduction of the complete method of Zyss et al.

Synthetic cell videos can be simulated [5–7] or generated by learned models [8–11]. SynCellFactory (SCF) [12] combines fitted statistics, ControlNet and label refinement; ASIST [13] trains segmentation and tracking with simulations. We compare rendering with direct painting under matched training budgets to test whether visual complexity improves association.

3. METHOD

3.1. Exchange-invariant mother–daughter scoring

Real-trained Trackastra [1] assigns link scores w_{pc} between consecutive observed frames. Candidate construction uses score threshold .05, distance at most 256 pixels and at most ten spatial neighbors per source node. All daughter pairs among retained outgoing edges are scored; ground-truth links are not inserted at inference.

A frozen ResNet-18 classifier encodes three-frame cell crops. At $8\times$, an interior crop uses $t-8, t, t+8$, with repeated endpoints and no hidden frames. Shared projections map 256-dimensional embeddings to 32 dimensions. An MLP combines the mother vector, symmetric daughter features (mean, absolute difference, product), mother–mean comparisons and 25 geometry/context values. All inputs are exchange-invariant; $q_p(a, b)$ is clipped to $[-4, 4]$. The public specification lists every feature and normalization.

3.2. Joint selection of divisions and continuations

Each parent p selects one child set S, and each child has at most one parent. We maximize the sum of

$$U_p(S) = \begin{cases} 0, & S = \emptyset, \\ \sum_{c \in S} w_{pc} + \beta \bar{q}_p(S) + .75(2 - |S|), & 1 \le |S| \le 3. \end{cases} \tag{1}$$

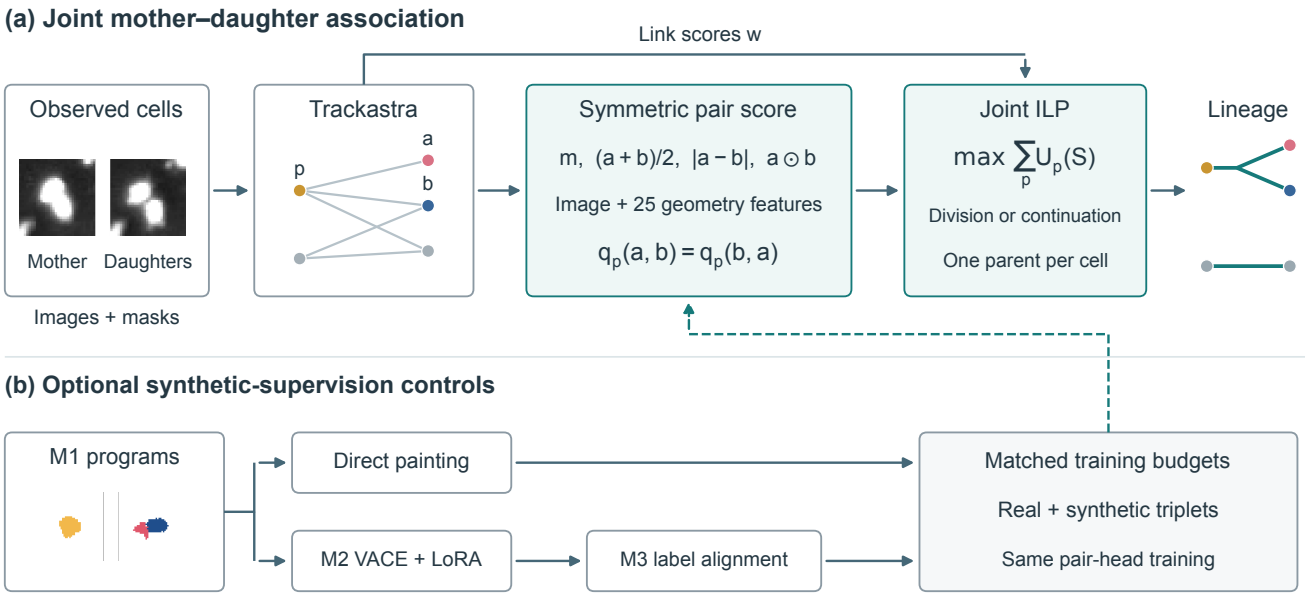


Fig. 1. MitoSync. (a) The core real-only tracker scores mother–daughter triplets and jointly selects competing continuations and divisions. The pair score is invariant to daughter order; the optional parent head is evaluated separately. (b) Synthetic-supervision controls share programmed events and compare direct painting with video rendering and M3 label alignment. Dashed arrows denote training supervision. Micrographs are recorded samples; candidate graphs illustrate the optimization, not measured predictions.

Here $\bar{q}_p$ averages unordered pair scores and is zero for a singleton, avoiding more pair terms merely from a larger set. The .75 coefficient inherits Trackastra `deepcell_gt` costs. The cap of three accommodates annotation branches, not three-way mitosis. These are fixed conventions, not individually optimized choices.

ILP selects consistent sets across each transition. With $\beta = 0$ it gives the graph reference. Learned-score configurations require a nonempty set for parents linked in that reference. This continuity constraint uses observed predictions, not test lineage labels. We test its removal with scores fixed.

3.3. Training and controlled variants

The pair head uses 6000 AdamW updates, batch 64 and balanced binary cross-entropy. Batches balance gaps 1/2/4/8 and positive/negative triplets; hard negatives use the four nearest daughters. R uses real data; RG mixes real/rendered data equally. Five seeds vary initialization, sampling and Trackastra checkpoints; corpora and the real-pretrained encoder remain fixed.

Validation selects $\beta = .125/.25$ for PSC at $1\times/8\times$; Fluo uses .5/.25, with only five native validation divisions. The Full control adds αr_{pc} per link, a clipped parent-versus-entry logit with $\alpha = .25$. Pair uses $\alpha = 0$. Full is retained to isolate earlier synthetic-data effects, not to define the core tracker.

3.4. Programmable supervision as a controlled alternative

M1 generates 17-frame identity masks and lineage links from training-fitted cycle, growth, shape and motion statistics. Initial ages follow a growing population [14]; the mother axis biases division [15]. Daughter separation, non-overlap and entry are constrained. Rendered events can deviate from the program.

M2 adapts Wan2.1-VACE-1.3B [16,17] with rank-32 LoRA [18] on 15 blocks, using latent flow matching and 113 PSC/90 Fluo windows. Masks provide control and a real frame supplies appearance. Generation uses 30 steps and guidance 5. To account for four-frame VAE groups, births shift to calibrated phases $\phi = (t_b - 1) \bmod 4$: $\{1, 2\}$ for PSC and $\{0, 2\}$ for Fluo.

M3 aligns labels with frozen Cellpose-SAM [19] detections by one-to-one mask matching at IoU $\geq .5$. The detected birth is

$$t^* = \min\{t \geq t_b : \text{both daughters match at } t \text{ and } t+1\}. \quad (2)$$

M3 keeps on-time births, re-times delays, and removes unmatched splits by assigning daughter regions to the mother identity. PSC yields 1085 on-time, 2435 re-timed and 2527 dropped events from 6047 programs. The same rule detects 63% of real divisions on time. Acceptance requires 80% matched events and PSC instance F1 $\geq .6986$. After two retries, fallback selects the highest-match attempt passing the instance gate (176/191 PSC clips); retained events still satisfy Eq. (2). This checks detection consistency, not ancestry.

4. EXPERIMENTS AND RESULTS

4.1. Data, protocol and comparisons

PSC and Fluo-N2DL-HeLa (Fluo) each have two annotated sequences [20,21]. Video 02 provides training frames 0–254/0–77 and validation tails of 45/14 frames. Video 01 has 300/92 frames and 569/94 divisions. It informed earlier development and the decision to reselect Fluo native weight on validation. CTC evaluation is retrospective; no hidden-test ranking is claimed.

CTC methods share silver-truth detections. We report zero-tolerance division F1 (BC(0)) and complete tracks (CT) [20,22]. At $8\times$, frames $o + 8j$ are retained for offsets $o = 0, \ldots, 7$. Unobserved tracks are removed; surviving endpoints are remapped, retaining parent links only when the parent survives. No missing ancestor is substituted. A division requires the mother and at least two daughters. Views average 515.4/86.0 eligible PSC/Fluo divisions. Public

Table 1. Video 01 F1 (BC(0)) and complete tracks (CT), mean±SD; bold marks maxima. Greedy: released linker; Graph: $\alpha = \beta = 0$; Pair: the core model; Full: Pair plus the optional parent head. Detections are shared; greedy may omit isolated cells.

	PSC 1×		PSC 8×		Fluo 1×		Fluo 8×	
Method	F1	CT	F1	CT	F1	CT	F1	CT
LapTrack	.730	.556	.589	.518	.856	.764	.760	.683
MAGIK (uniform)	.770±.005	.536±.007	.454±.019	.377±.008	.710±.019	.608±.019	.616±.035	.512±.024
+ MitoSync	.797±.010	.571±.011	.472±.026	.392±.014	.729±.063	.642±.036	.500±.070	.456±.050
Trackastra greedy	.837±.007	.651±.010	.696±.007	.592±.007	.943±.009	.861±.015	.854±.024	.787±.021
Trackastra graph	.857±.008	.660±.008	.696±.007	.594±.010	.947±.010	.871±.017	.830±.038	.767±.032
Pair (real only)	.874±.007	.670±.009	.749±.004	.656±.007	**.954**±.003	**.884**±.005	.909±.007	.848±.004
Pair + MitoSync	**.888**±.006	**.682**±.004	**.755**±.003	.667±.004	.950±.002	.880±.005	**.913**±.007	**.854**±.004
Full (real only)	.862±.006	.671±.006	.741±.003	.669±.004	**.954**±.003	**.884**±.005	.899±.008	.839±.005
Full + SCF-adapted	.857±.009	.665±.009	.731±.005	.660±.004	.946±.007	.880±.005	.891±.006	.831±.003
Full + MitoSync	.879±.003	.679±.002	.749±.005	**.674**±.004	.950±.002	**.884**±.006	.909±.006	.846±.003

records also report BC(1), BC(2), TRA, DET and LNK; tolerance is measured in observed frames.

Offsets are averaged within seed. Seed variability measures training repeatability, not cross-video uncertainty. Sparse Pair-versus-graph F1 is the main reported endpoint, chosen after development rather than preregistered. All inference is exploratory. The public statistics map preserves the original comparison families and adjusted intervals; none is replaced to change significance.

LapTrack [3] uses parameters fitted on all video 02. MAGIK [2] uses geometry inputs and three seeds; other learned trackers use five. The SCF-adapted control uses released SCF code and training schedules [12], but generic Cellpose-SAM replaces its fine-tuned Cellpose while simulated-position correction is retained. It is therefore an adapted pipeline comparison, not a reproduction of the original segmentation stage. CTC conclusions concern association under shared detections, not end-to-end superiority over all CTC trackers.

4.2. Association and decoding

The real-only Pair model improves sparse F1 over graph by 5.24 PSC and 7.95 Fluo points (Table 1) and exceeds the released greedy linker. The gain persists at one- and two-observation division tolerances. It matches Full+MitoSync in rounded sparse F1. Parent-only R scores are lower than graph in all settings; removing parent scores from RG does not clearly change F1 after correction.

Removing continuity with scores fixed changes sparse R/RG F1 by at most 0.11 points. Native PSC is unchanged; native Fluo loses 1.05/1.32 R/RG points. Sparse gains remain.

Joint daughter scores provide the largest component gain (Table 2). Zeroing image embeddings and retraining reduces Full F1 by 1.25–1.71 R points and 0.64–1.20 RG points on PSC and sparse Fluo; native Fluo effects are not established.

4.3. Transfer and fixed-score decoder controls

Frozen Fluo models transfer without retraining or tuning to 36 LFCT fields from four cell lines [23], using shared predicted Cellpose-SAM masks. Of 53 requested fields, 11 have multi-parent annotations, five fail input checks and one is unavailable. Shared exclusions are individually recorded. Exact triplet event F1 uses centroid-in-mask assignment and differs from CTC BC(0).

Sparse R improves F1 over graph by 4.05 points (adjusted interval: 2.38–5.25), with positive means in all four cell lines. Native means are lower and adjusted intervals include zero (Table 3).

Table 2. Component F1 over five seeds; bold marks maxima. No pair: no daughter scores; Geometry pair: zeroed image embeddings; Full: both heads. R/RG: real/real+synthetic.

		PSC 01		Fluo 01	
Stage	Data	1×	8×	1×	8×
No pair	R	.838±.008	.693±.006	.909±.011	.820±.015
	RG	.866±.006	.701±.009	.928±.006	.859±.015
Geometry pair	R	.846±.006	.724±.004	.950±.003	.886±.011
	RG	.872±.004	.737±.006	.950±.002	.898±.007
Full	R	.862±.006	.741±.003	**.954**±.003	.899±.008
	RG	**.879**±.003	**.749**±.005	.950±.002	**.909**±.006

Table 3. LFCT event F1, averaged over offsets, seeds and fields. Lower rows keep the same frozen pair scores and utility; bold marks block maxima. Continuity means retaining a nonempty child set for parents linked by the graph reference.

Configuration	Native	Sparse
Trackastra graph	**.140**	.098
Pair R, continuity	.130	.138
Pair RG, continuity	.126	**.141**
Pair R, ILP without continuity	**.127**	.141
Pair R, greedy without continuity	.122	.126
Pair RG, ILP without continuity	.123	**.144**

Field-bootstrap intervals describe this cohort; biological independence is unverified. At native and sparse rates, R endpoint coverage is 54.2% and 57.5%; precision is 11.2% and 15.0%. Missing endpoints, missed links and false branches limit absolute F1, precluding reliable lineage measurement.

With R scores fixed and continuity disabled, ILP exceeds greedy by 0.45 native and 1.45 sparse points (adjusted intervals [0.21,0.75] and [0.82,2.17]). Continuity reduces sparse R/RG by 0.25/0.29 points. This isolates solver effects under one utility, not superiority over the complete method of Zyss et al. [4].

4.4. What does synthetic supervision add?

Generation accepts 191/584 PSC and 200/352 Fluo attempts, including retries. Labels contain 3520/3554 divisions versus 260/202 real training divisions. Fluo counts include frame-0 mothers; excluding

Table 4. Five-seed division F1; bold marks block maxima. Raw: rendered images/program labels; RG/PM: rendered/painted images with corrected labels. Lower block: original-label painting versus rendering/M3; matched events, pair-only decoding.

	PSC 01		Fluo 01	
Data	1×	8×	1×	8×
Full tracker; original sampling				
Raw	.869±.006	.745±.005	.948±.005	.908±.006
RG	.879±.003	.749±.005	**.950**±.002	**.909**±.006
PM	**.885**±.005	**.750**±.004	**.950**±.002	**.909**±.007
Pair-only; matched events				
Direct paint	.884±.006	**.754**±.002	.946±.007	**.916**±.005
Render+M3	**.888**±.006	**.754**±.003	**.950**±.002	.913±.006

them for classification gives 3424/199 without changing the corpus. Figure 2 shows selected event-rich clips, not estimates of real population density.

With Full fixed, synthetic training adds 1.70 native PSC, 0.79 sparse PSC and 0.99 sparse Fluo F1 points; native Fluo loses 0.39 points. On identical rendered images, M3 label correction improves PSC F1 by 0.94/0.35 points but has no clear Fluo effect. Painting the corrected masks (PM) remains competitive with rendering (RG), so image realism is not an established source of benefit.

Direct painting bypasses rendering and M3, using original M1 labels. Both arms match events, negative pools, source ratios, updates and seeds. Rendering adds 0.44 native PSC points (paired 95% interval [0.27,0.62]), surviving Holm correction; other differences are not established. Neither arm differs clearly from real-only in the matched eight-comparison family. LFCT rendering trails painting by 0.41/0.54 native/sparse points. A general rendering advantage is unproven; nonsignificance is not equivalence.

MitoSync exceeds SCF-adapted by 1.19/2.37 sparse PSC/Fluo points with training budgets matched, but segmentation adaptation and generation cost differ. Uniform-window MAGIK loses 11.6 sparse Fluo points (Table 1). Separate balanced sampling gives R/RG .680/.755 native and .616/.630 sparse, with both difference intervals including zero. Its .680 baseline uses a different protocol; cross-tracker benefit is not established.

Additional M1 dynamics and production phase scheduling do not establish downstream gains after correction. Scheduling raises first-attempt on-time matches from 140/1283 to 196/1310 PSC events and 295/868 to 348/872 Fluo events. M3 can then re-time labels, separating timing improvement from training utility.

4.5. Label review and computational cost

One author, blinded to M3 decisions, assessed 72 PSC events from 67 clips in density-by-decision strata. Weighted visible-division rates are 95.9% on-time, 66.5% re-timed and 41.2% dropped. The retained rate is 75.6% (clip-clustered interval [61.6,88.0]), including uncertain/lost cases; its complement is not a confirmed error rate. The 1.32-frame onset difference compares human separation with M3 persistence.

Clicks match training-ID timing in 17/39 retained positives; 22 have conflicts. In eight conflicts checked again at frames 0/16, seven triplets agree and one maps both daughters to one ID. This targeted check does not estimate corpus accuracy; inter-rater agreement is unmeasured. Masking re-timed/dropped families discards useful samples without clear gains over random removal.

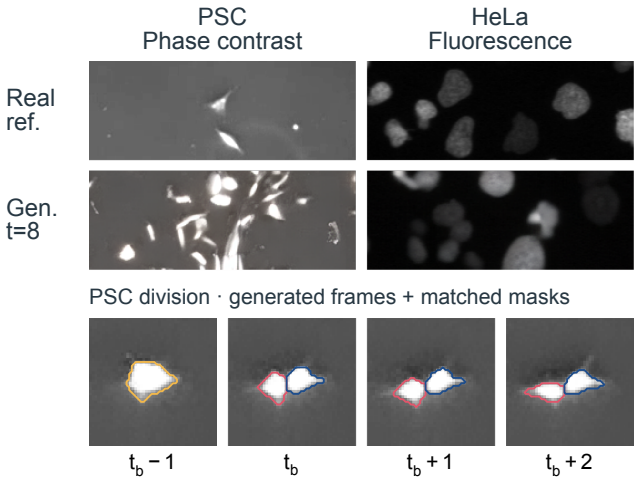


Fig. 2. Real appearance references and generated PSC/Fluo images, with consecutive 48×48-pixel PSC event crops below. A reference supplies appearance, not a spatially corresponding frame. These selected examples do not measure realism; generated density need not match the reference. Time labels count generated frames, not calibrated biological time.

PSC/Fluo job durations sum to 5.41/5.40 hours for LoRA, 5.12/3.97 for generation and 2.35/1.96 for readout. Painting takes 7.49/7.69 CPU minutes. Generation used concurrent 96 GB RTX PRO 6000 GPUs; incomplete hardware/concurrency records preclude GPU-hour or end-to-end cost ratios.

5. DISCUSSION AND CONCLUSION

MitoSync improves sparse association by scoring mother–daughter candidates jointly and optimizing competing cell assignments. Real-only results and fixed-score controls support this contribution. External sparse gains extend evidence beyond development videos. Native-rate gains and practical lineage recovery need validation.

Synthetic supervision has setting-dependent benefits; direct painting remains competitive at lower processing cost. Detector consistency does not guarantee lineage labels. Future validation needs untouched videos, broader predicted-mask benchmarks and multiple reviewers. These results distinguish association accuracy from supervision quality.

Code, configurations, evaluation data, method definitions and generated examples are public at `https://github.com/CHEN-Yunshu/MitoSync-ISBI2027`. Full synthetic corpora and weights are not included.

6. COMPLIANCE WITH ETHICAL STANDARDS

We used public Cell Tracking Challenge [21] and LFCT [23] data. No new human or animal data were collected, so ethical approval was not required. The authors have no relevant financial or non-financial interests to disclose.

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